



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
20-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Ishan wants to float a ball (perfectly reflecting) in the air. He turned on a monochromatic light source at the bottom, perpendicular to the ground, pointing upwards with intensity P . The ball is dropped from a height H , directly into the light beam. The ball has a radius R and a mass m . Unfortunately, Ishan did not compute the needed intensity correctly, and the light beam cannot keep it floating in the air. At what speed does the ball hit the ground? (Symbols have their usual meaning.)

(A) $\sqrt{H \left(g - \frac{\pi R^2 P}{mc} \right)}$

(B) $\sqrt{2H \left(g - \frac{\pi R^2 P}{2mc} \right)}$

(C) $2\sqrt{H \left(g - \frac{\pi R^2 P}{mc} \right)}$

(D) $\sqrt{2H \left(g - \frac{\pi R^2 P}{mc} \right)}$

2. Gill has a dry well with constant circular cross-section and depth h . When standing at the bottom, he is able to see an angle $2\alpha \leq 90^\circ$ of the sky above him. We want to increase this angle by pouring some specific liquid into the well (Gill still stay at the bottom). Find the condition for the index of refraction of the liquid if we want the angle of the visible sky to be twice as large.

(A) $n \geq 2 \cos \alpha$

(B) $n \leq 2 \cos \alpha$

(C) $n \geq 2 \sin \alpha$

(D) $n \leq 2 \sin \alpha$

3. Rohit wants to find out how the refractive index of the atmosphere should depend on the height y above the Earth's surface, if rays in the atmosphere were to travel along parabolic trajectories $y = x^2$. The dependence of refractive index on y may be (You may assume earth to be flat):
- (A) $n = n_0 \sqrt{2y + 1}$
(B) $n = n_0 \sqrt{4y + 1}$
(C) $n = n_0 \sqrt{16y + 1}$
(D) None
4. The energy–momentum relation is the relativistic equation relating total energy (which is also called relativistic energy) to invariant mass (which is also called rest mass) and momentum. It is the extension of mass–energy equivalence for bodies or systems with nonzero momentum. It can be written as the following equation $E = \sqrt{(m_0 c^2)^2 + (pc)^2} = mc^2$. This equation holds for a body or system, such as one or more particles, with total energy E , invariant mass m_0 , and momentum of magnitude p ; the constant c is the speed of light. X-ray photons are scattered through an angle of 90° by electrons initially at rest. What is the change in the wavelength of the photons? (Take rest mass of electron as m_0)
- (A) $\frac{h}{\sqrt{2}m_0 c}$
(B) $\frac{\sqrt{2}h}{m_0 c}$
(C) $\frac{h}{m_0 c}$
(D) None

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

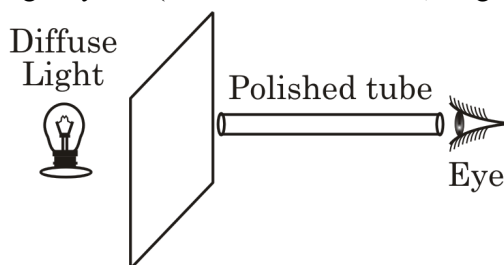
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. You are given a long, thin polished metal tube and asked to put your eye at one end and to point the other end at a diffuse light source (a well-illuminated wall, for example). What you see is a pattern of rings. At the centre there is a bright circle, which is the diffuse light source seen directly through the hole in the end of the tube. Surrounding this is a ring of light that is less bright than the central circle. Around that there's another ring that's less bright still. And so on. The rings become successively less bright until the pipe appears dark. The rings appear to have approximately the same width as they get larger. The angle subtended at the eye by ring may be : (Diameter of tube is D, length of tube is L)



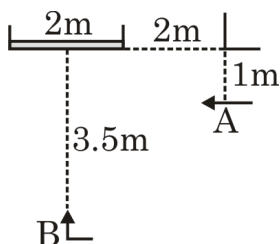
(A) $\frac{D}{2L}$

(B) $\frac{D}{L}$

(C) $\frac{3D}{2L}$

(D) $\frac{2D}{L}$

6. What atomic number would an element have to have for the ground state speed of an electron to exceed the speed of light, if we do not consider relativistic corrections? (Use the Bohr model of the atom and assume that the atom is ionized to exactly one electron.)
- (A) 108 (B) 118
(C) 128 (D) 138
7. On a big room's wall there is a mirror of width 2 m. A person is standing 2 m away from the mirror and 1 m away from the wall. The person starts to move parallel to the mirror with a speed of 1 m/s. At the same moment the person's acquaintance starts to move with a speed of 1 m/s towards the mirror on the mirror's centerline. Initially, the acquaintance is standing at a distance 3.5 m from the mirror. After what time do the two see each other from the mirror?



- (A) 1 sec (B) 1.5 sec
(C) 2 sec (D) 2.5 sec

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. List-I contain hydrogen like atom/ion and List-II contains their properties according to Bohr's model. Select the correct match.

	List-I		List-II
(P)	H, electron in ground state.	(1)	Number of revolutions per seconds is $\frac{me^4}{\epsilon_0^2 h^3}$.
(Q)	He ⁺ , electron in ground state.	(2)	Radius of orbit is $\frac{\epsilon_0 h^2}{\pi m e^2}$.
(R)	Li ²⁺ , electron in 3 rd orbit.	(3)	Velocity of electron is $\frac{e^2}{\epsilon_0 h}$.
(S)	Be ³⁺ , electron in 2 nd orbit.	(4)	Kinetic energy of electron is $\frac{me^4}{8\epsilon_0^2 h^2}$.
		(5)	Potential energy is $-\frac{me^4}{\epsilon_0^2 h^2}$.

- (A) P → 4; Q → 2; R → 1; S → 3
 (B) P → 2; Q → 4; R → 1; S → 5
 (C) P → 2; Q → 3; R → 4; S → 5
 (D) P → 2; Q → 4; R → 2; S → 3

9. Select the correct match.

	List-I		List-II
(P)	The filament voltage in X-ray experiment is increased.	(1)	The average kinetic energy of the electrons decreases.
(Q)	In photoelectric effect work function of the target is increased.	(2)	The average kinetic energy of the electron increases.
(R)	The de-Broglie wavelength of β -rays is increased.	(3)	The intensity increases.
(S)	The de-Broglie wavelength of β -rays is decreases.	(4)	The cut-off threshold wavelength decreases.
		(5)	The cut-off threshold wavelength remains unchanged.

(A) $P \rightarrow 3,5; Q \rightarrow 1,4; R \rightarrow 1; S \rightarrow 2$

(B) $P \rightarrow 2,4; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 3,5; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

(D) $P \rightarrow 3,4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

10. Select the matching list.

	List-I		List-II
(P)	Convex mirror, virtual object.	(1)	Real image.
(Q)	Concave mirror, virtual object.	(2)	Virtual image.
(R)	Concave lens, real object.	(3)	Enlarged image
(S)	Convex lens, real object.	(4)	Diminished image
		(5)	Positive focal length.

(A) $P \rightarrow 1,2,3,4,5; Q \rightarrow 1,4; R \rightarrow 2,4; S \rightarrow 1,2,3,4,5$

(B) $P \rightarrow 1,2,5; Q \rightarrow 1,3,4; R \rightarrow 2,4; S \rightarrow 1,2,5$

(C) $P \rightarrow 1,2,4,5; Q \rightarrow 1,3,4; R \rightarrow 2,3,4; S \rightarrow 1,2,3,4,5$

(D) $P \rightarrow 1,2,3,4,5; Q \rightarrow 1,3,4; R \rightarrow 2,4; S \rightarrow 1,2,3,4,5$

11. Select the matching list.

	List-I		List-II
(P)	Mosley's law	(1)	Characteristic x-ray.
(Q)	Change of photon energy into kinetic energy of electrons	(2)	Photoelectric effect.
(R)	Electron emission from a material.	(3)	Hydrogen emission spectrum.
(S)	Transition between two atomic energy levels.	(4)	γ -decay
		(5)	α -decay (only)

- (A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 2$
- (B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$
- (C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 2$
- (D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 1, 3$

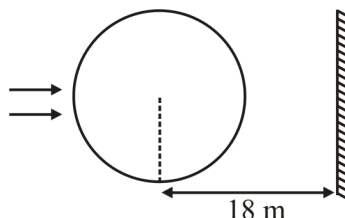
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

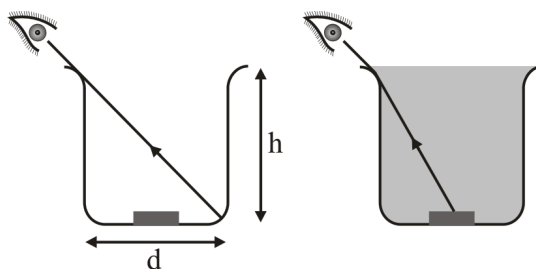
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

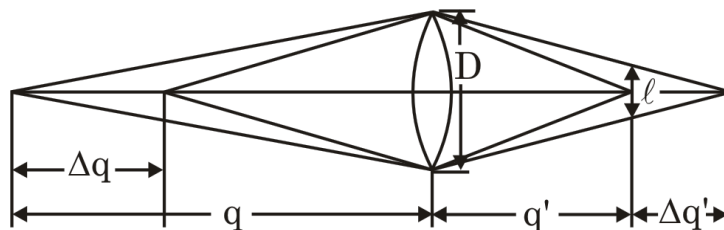
1. Virat got a very large number of really thin convex lenses for Diwali. He placed them very closely one behind the other so that the first one had a focal length $f = 10\text{cm}$, the second one $2f$, a third $4f$ and so on. He also found a concave mirror with radius of curvature f , which he placed right behind this row of lenses. He inserted a luminous object into the focus of the first lens. What distance (in cm) in front of the mirror did its image appear?
2. A metallic surface is illuminated alternatively with electromagnetic radiations of wavelengths 3000 Å and 6000 Å . It is observed that the maximum speeds of the photo electrons under these illuminations are in the ratio $3 : 1$. Calculate the work function of the metal (in eV). Round off to nearest integer.
3. A hydrogen atom with its electron in the n th orbit is ionized by electromagnetic radiation of wavelength 45 nm . If the kinetic energy of the ejected electron is 26.75 eV then the value of n is ($hc = 1242\text{ eV nm}$).
4. A parallel narrow beam of light is incident on a glass sphere of radius R and index 1.5 as shown in figure. A plane mirror is at distance 18m from the centre of sphere. Find the distance of final image from the plane mirror. (Take $R = 8\text{ m}$)



5. A person looking into an empty container is able to see the far edge of the container's bottom as shown. The height of the container is h , and its width is d . When the container is completely filled with a fluid of index of refraction n and viewed from the same angle, the person can see the center of a coin at the middle of the container's bottom as shown. Find $(h/d)^2$ if $n = \sqrt{3}$.



6. For a camera lens, "depth of field" is how far a point object can be away from the position where it would be precisely in focus and still have the light from it fall on the film within a "circle of confusion" of some diameter, say ℓ . For a given picture Δq is depth of field, the focal length of the lens is f and D is diameter of lens. Find depth of field (in m). (You may consider the object distance to be much larger than the focal length.) ($\ell = 1 \text{ mm}$, $D = 5 \text{ cm}$, $q = 5 \text{ m}$, $f = 50 \text{ cm}$)



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Morphine changes into Heroin by replacing

- (A) Both phenolic and alcoholic $-\text{OH}$ by $-\text{OCH}_3$
 (B) Phenolic $-\text{OH}$ by $-\text{OCH}_3$
 (C) Both phenolic and alcoholic $-\text{OH}$ by $-\text{OAc}$
 (D) Alcoholic $-\text{OH}$ by $-\text{OCH}_3$

2.
$$\left[\text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_2 - \overset{\text{CN}}{\underset{|}{\text{CH}}} \right]_n$$

The name of polymer is :

- (A) Polystyrene (B) BuNa-S
 (C) BuNa-N (D) BuNa-CN

3. If DNA fragment is $3' \text{---} \overline{\text{A T G A C T G G A}} \text{---} 5'$ then it's transcribed RNA fragment is :

- (A) $3' \text{---} \overline{\text{A T G C T C C A T}} \text{---} 5'$ (B) $3' \text{---} \overline{\text{U A C U G A C C U}} \text{---} 5'$
 (C) $5' \text{---} \overline{\text{U A C U G A C C U}} \text{---} 3'$ (D) $3' \text{---} \overline{\text{U T C U G U C C U}} \text{---} 5'$

4. Hydrolysis of sucrose into (+) glucose and (-) fructose is known as-

- (A) Mutarotation (B) Inversion
 (C) Pyrolysis (D) None of these

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

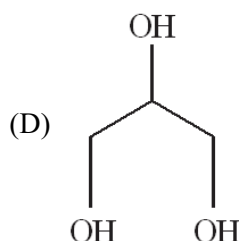
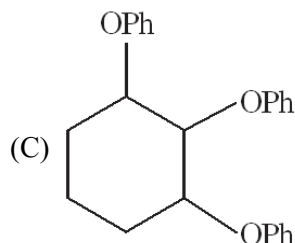
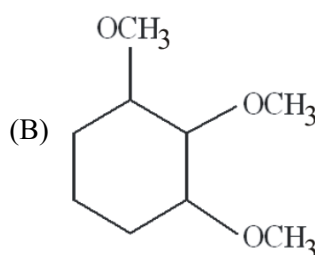
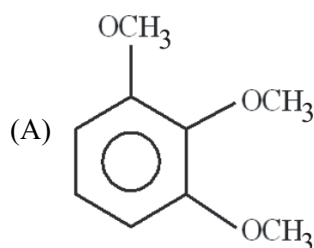
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. One mole of the following compound(s) can consume more than 4 moles of HI.



6. Which reagent can be used to distinguish t-Butyl alcohol and isopropyl alcohol :
- (A) KMnO_4
 - (B) $\text{HCl} + \text{Anhy ZnCl}_2$
 - (C) Ammonical AgNO_3
 - (D) $\text{K}_2\text{Cr}_2\text{O}_7$
7. Following statement(s) is/are true for D-glucose :
- (A) It shows Schiff's reagent test
 - (B) It is C-2 epimer of D-mannose
 - (C) Oxidise with $\text{Br}_2 / \text{H}_2\text{O}$ to form saccharic acid
 - (D) Shows mutarotation.

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8.

List-I		List-II	
(P)	Ethinylestradiol	(1)	Analgesic
(Q)	Salvarsan	(2)	Tranquilisers
(R)	Equanil	(3)	Antibiotics
(S)	Paracetamol	(4)	Antifertility drug
		(5)	Antipyretic

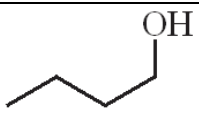
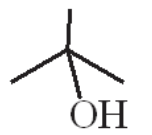
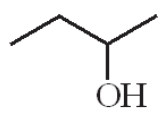
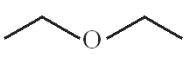
(A) $P \rightarrow 1; Q \rightarrow 2,5; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2,5; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1,5$

(D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1,5$

9. List-I contains four isomers of $C_4H_{10}O$ and List-II contains reactions & properties.

	List-I		List-II
(P)		(1)	Produces H_2 on reaction with metallic Na
(Q)		(2)	Undergoes oxidation to aldehyde or ketone by PCC
(R)		(3)	Cloudiness appears immediately in Lucas Test at room temperature
(S)		(4)	Reaction with conc. HI it gives mixture of ethyl alcohol and ethyl iodide
		(5)	Gives red color in Victor Meyer's test.

- (A) $P \rightarrow 1,2,5; Q \rightarrow 1,3; R \rightarrow 1,2; S \rightarrow 4$
 (B) $P \rightarrow 1,2,3; Q \rightarrow 1,3; R \rightarrow 1,3; S \rightarrow 5$
 (C) $P \rightarrow 2,3,4; Q \rightarrow 1,4; R \rightarrow 2,3; S \rightarrow 4,5$
 (D) $P \rightarrow 1,4; Q \rightarrow 2,5; R \rightarrow 2,4,5; S \rightarrow 1,2$

10. Match the List I with List II and select the correct answer for major product using the codes given below the Lists.

List-I		List-II	
(P)	$\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{Br} \xrightarrow[\Delta]{\text{C}_2\text{H}_5\text{O}^- \text{Na}^+}$	(1)	Elimination Reaction
(Q)	$\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{OH} \xrightarrow{\text{H}^+ / \Delta}$	(2)	Carbocation intermediate
(R)	$\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{Br} \xrightarrow{\text{H}_2\text{O}}$	(3)	Reaction follows 1 st order kinetics
(S)	$\text{CH}_3 - \text{Br} \xrightarrow{\text{CH}_3\text{CH}_2\text{O}^- \text{Na}^+}$	(4)	Reaction follows 2 nd order kinetics
		(5)	Reaction is solvolysis

The correct option is

- (A) P → 1,2; Q → 1,2,3; R → 2,3; S → 4
 (B) P → 1,2,4; Q → 1,2,3; R → 2,3; S → 3,4
 (C) P → 1,2; Q → 1,3; R → 2,3; S → 4,5
 (D) P → 1,4; Q → 1,2,3; R → 2,3,5; S → 4

11. Match the list

List-I		List-II	
(P)	Acetophenone	(1)	Reaction with AgNO ₃ + NH ₄ OH to give white ppt.
(Q)	Benzaldehyde	(2)	Reaction with NaOI to give yellow ppt. of CHI ₃
(R)	Butanal	(3)	Reaction with Fehling to give brown ppt. of Cu ₂ O
(S)	1-butyne	(4)	Reaction with AC ₂ O + AcONa to give Cinnamic acid
		(5)	Reaction with Tollen's reagent give silver mirror

The correct code for your answer :-

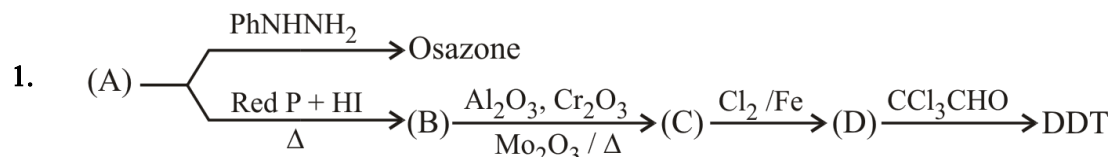
- (A) P → 2; Q → 4; R → 3; S → 1
 (B) P → 1; Q → 4; R → 3; S → 2
 (C) P → 2; Q → 3; R → 4; S → 1
 (D) P → 1,4; Q → 2; R → 4; S → 3

SECTION-II : (Maximum Marks: 24)

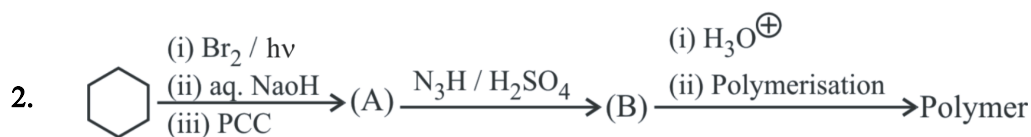
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

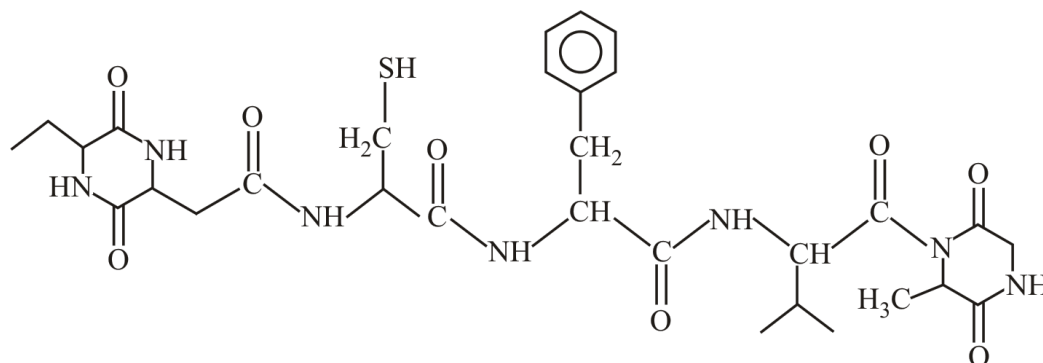


How many moles of HIO_4 will be consumed by one mole of compound (A)?

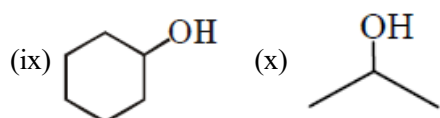
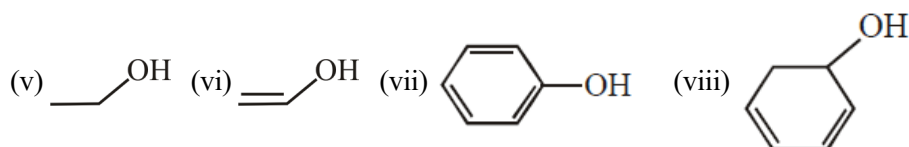
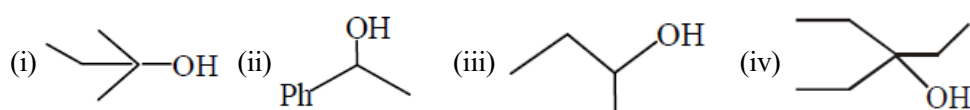


The degree of unsaturation of compound (B) is ____

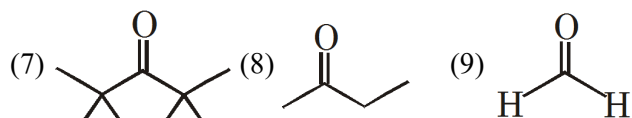
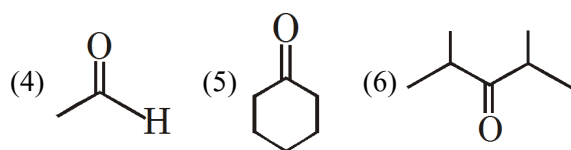
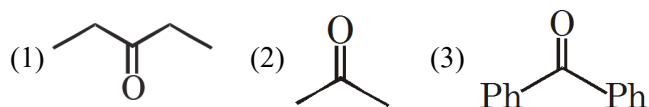
3. How many total natural and non essential Amino acids are obtain after complete acidic hydrolysis of following compound.



4. The alcohol among following which show dehydration reaction faster than 2-methyl propan-2-ol ().

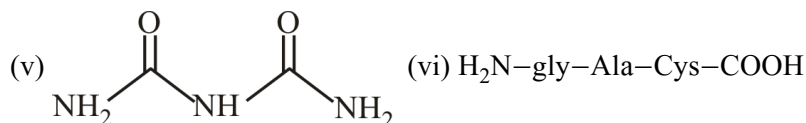


5. Identify compounds give positive test with NaHSO_3 ?



6. Identify compounds give Biurate test?

(i) Cellulose (ii) Protien (iii) Nylon-6 (iv) Starch



PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $f(x)$ is a polynomial function and $(f(\alpha))^2 + (f'(\alpha))^2 = 0$ then find $\lim_{x \rightarrow \alpha} \frac{f(x)}{f'(x)} \left[\frac{f'(x)}{f(x)} \right]$
(where $[.]$ denotes greatest integer function)
(A) 0 (B) 1
(C) -1 (D) 3
2. The function $f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$ is
(A) Continuous but not differentiable at $x = 1$
(B) Differentiable at $x = 1$
(C) Neither continuous nor differentiable at $x = 1$
(D) Differentiable everywhere
3. Range of the function $f(x) = \frac{2x^4 - 14x^2 + 8x + 49}{x^4 - 7x^2 - 4x + 23}$ is
(A) $[1, 3]$ (B) $[2, 3]$
(C) $(2, 3]$ (D) $(1, 2]$
4. If $f(\theta) = \frac{(2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 4\theta - 1) \dots (2 \cos 2^{n-1}\theta - 1)}{2 \cos 2^n \theta + 1}$ for $n \in \mathbb{N}$ and $\theta \neq 2m\pi \pm \frac{2\pi}{3}$,
 $m \in \mathbb{I}$ then $f\left(\frac{\pi}{4}\right)$ is equal to
(A) $\sqrt{2} - 1$ (B) $1 - \sqrt{2}$
(C) $\sqrt{3} - 1$ (D) $1 - \sqrt{3}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Suppose that $f(a) > 0$ and that f is differentiable at $x = a$. Which of the following statements is/are TRUE.

(A) $\lim_{n \rightarrow \infty} \left(\frac{f\left(a + \frac{1}{n}\right)}{f(a)} \right)^{\frac{1}{n}} = 1$

(B) $\lim_{n \rightarrow \infty} \left(\frac{f\left(a + \frac{1}{n}\right)}{f(a)} \right)^{\frac{1}{n}} = e^{\frac{f'(a)}{f(a)}}$

(C) $\lim_{x \rightarrow a^+} \left(\frac{f(x)}{f(a)} \right)^{\frac{1}{\ln x - \ln a}} = e^{\frac{af'(a)}{f(a)}}; a > 0$

(D) $\lim_{x \rightarrow a^-} \left(\frac{f(x)}{f(a)} \right)^{\frac{1}{\ln x - \ln a}} = e^{\frac{af'(a)}{f(a)}}; a > 0$

6. Which of the following equations have no real solutions ?

(A) $x^2 - 2x + 5 + \pi^x = 0$

(B) $\log_{1.5}(\cot^{-1}x + \operatorname{sgn}(e^x)) = 0$

(C) $x^4 - 2x^2 \sin^2 \frac{\pi x}{2} + 1 = 0$

(D) $\tan\left(x + \frac{\pi}{6}\right) = 2 \tan x$

7. Function $f(x)$ satisfies the relation $f(x) + 2f(1 - x) = x^2 + 1 \quad \forall x \in \mathbb{R}$ then :

(A) $f(x)$ is an odd function

(B) $f(x)$ is many-one function

(C) Range of $f(x)$ is $\left[-\frac{1}{3}, \infty\right)$

(D) $f(x)$ is increasing in the interval $(2, \infty)$

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. If $f(x)$ is twice differentiable function in $[c_1-1, c_2+1]$ and $f'(c_1) = f'(c_2) = 0$, $f'(c_1).f'(c_2) < 0$, $f(c_1) = 9$, $f(c_2) = 0$. Let k and m be the minimum number of the roots of $f(x) = 0$ and $f'(x) = 0$ in $[c_1-1, c_2+1]$.

List-I		List-II	
(P)	If $f'(c_1) - f'(c_2) > 0$, then $k =$	(1)	1
(Q)	If $f'(c_1) - f'(c_2) < 0$, then $k =$	(2)	2
(R)	If $f'(c_1) - f'(c_2) > 0$, then $m =$	(3)	3
(S)	If $f'(c_1) - f'(c_2) < 0$, then $m =$	(4)	4
		(5)	7

- (A) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 3$
- (B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 1$
- (C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$
- (D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 4$

9. α, β are real and distinct roots of $x^2 + 4bx + c = 0$ where $b, c \in \mathbb{R}$.

List-I		List-II	
(P)	$f(x) = x^3 + 6bx^2 + 3cx + 2;$ $b < 0, c > 0, f(\alpha)f(\beta) > 0$	(1)	$f(x) = 0$ has three real roots out of which two are positive real roots and one is negative real root
(Q)	$f(x) = cx^3 + 6bx^2 + 3x - 2;$ $c < 0; f(1/\alpha)f(1/\beta) > 0$	(2)	$f(x) = 0$ has exactly one real root which is positive
(R)	$f(x) = x^3 + 6x^2 + 3cx - 2;$ $c < 0, f(\alpha)f(\beta) > 0$	(3)	$f(x) = 0$ has three real roots out of which two are negative real roots and one is positive real root
(S)	$f(x) = cx^3 + 6bx^2 + 3x - 2;$ $c < 0, f(1/\alpha)f(1/\beta) < 0$	(4)	$f(x) = 0$ has exactly one real root which is negative
		(5)	$f(x) = 0$ has three positive real roots

(A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 3$

(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(C) $P \rightarrow 4; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

10. Given $f_1(x) = x^2 \operatorname{sgn}(x) = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$; $f_2(x) = \int_0^x t^2 \sin\left(\frac{1}{t}\right) dt$; $f_2(0) = 0$; $f_3(x) = x^{\frac{-1}{3}} |\sin x|$, $f_3(0) = 0$;
 $f_4(x) = \begin{cases} 0 & -1 < x \leq 0 \\ -x^3 & 0 < x < 1 \end{cases}$.

List-I		List-II	
(P)	The function f_1 is	(1)	continuous but not differentiable at $x = 0$
(Q)	The function f_2 is	(2)	first derivative exists at $x = 0$ but second derivative does not exist
(R)	The function f_3 is	(3)	second derivative exists at $x = 0$, but it is not continuous there
(S)	The function f_4 is	(4)	second derivative exists at $x = 0$ and is continuous
		(5)	Discontinuous at $x = 0$

The correct option is :

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

(D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

11. If $f(x) = \sin^{-1} \left(\frac{4x}{4+x^2} \right)$, then

List-I		List-II	
(P)	$f(\sqrt{8}-2)$	(1)	$\pi + \tan^{-1} \left(\frac{4\pi}{4-\pi^2} \right)$
(Q)	$f \left(-\frac{3}{\sqrt{8}-2} \right)$	(2)	$-\pi + 2\tan^{-1} \left(\frac{3}{4}(\sqrt{2}+1) \right)$
(R)	$f(\sqrt{12}-\sqrt{3})$	(3)	$\tan^{-1} \frac{4\pi}{\pi^2-4}$
(S)	$f(\pi)$	(4)	$\frac{\pi}{2} - 2\tan^{-1}(\sqrt{2}-1)$
		(5)	$\frac{\pi}{2} - \tan^{-1} \left(\frac{\sqrt{3}}{12} \right)$

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$
 (B) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$
 (C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 1$
 (D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

1. Find the number of real solution of the equation $x^3 + 3x^2 - (x - 6)^{1/3} + 8 + 3x = 0$
2. The number of solutions of the equation $\left(x - 2 \cdot \left[\frac{x}{2}\right] - 1\right)^2 = \frac{1}{2}$ for $x \in [-4, 4]$, is (where $[.]$ denotes greatest integer function)
3. Let $f : A \rightarrow \left[-\frac{1}{2}, 0\right]$ is such that $f^3(x) - 3xf(x) + x^3 + 1 = 0$ and $f(x)$ is an invertible function, then number of solution(s) of the equation $(f(f(x)))^2 = \frac{3}{2} f^{-1}(x) + 1$ is
4. Let $P(x)$ be a polynomial of degree 4 having a local maximum at $x = 2$ and $\lim_{x \rightarrow 0} \left(3 - \frac{P(x)}{x}\right) = 27$. If $P(1) = -9$ and $P''(x)$ has a local minimum at $x = 2$. If global maximum value of $y = P'(x)$ on the set $A = \{x : x^2 + 12 \leq 7x\}$ is $4M$ then value of M is :
5. Number of solution of the equation $\left(\cos^{-1} \left(\frac{1 - 4x^2}{1 + 4x^2}\right)\right)^{\frac{3}{4}} = x^2 - x \sin x - \cos x$ is
6. Let $f(x)$ is defined only for $x \in (0, 5)$ and defined as $f^2(x) = 1 \forall x \in (0, 5)$. Function $f(x)$ is continuous for all $x \in (0, 5) - \{1, 2, 3, 4\}$ (at $x = 1, 2, 3, 4$ $f(x)$ may or may not be continuous). The number of possible function $f(x)$ if it is discontinuous at two and three integral points in $(0, 5)$ are a and b respectively, then $a + b$ is equal to

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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